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Via email and U.S. Mail

Aimee Wilson
Air Permits Section (R6 PD-R)
U.S. EPA Region 6
1445 Ross Avenue, Suite 1200
Dallas, TX 75202
wilson.aimee@epa.gov

RE: Corpus Christi Liquefaction LNG Terminal–Permit No. PSD-TX-1306-GHG

Dear Ms. Wilson:

These comments are submitted on behalf of Sierra Club and its 600,000 members, including over 21,000 members in Texas. The issues addressed below regarding the proposed *Draft Prevention of Significant Deterioration Permit for Greenhouse Gas Emissions* for the Corpus Christi Liquefaction, LLC (Applicant) LNG terminal (CCL Project) are based off of publicly available materials, including the February 2014 Statement of Basis (SOB) prepared by EPA Region 6 (the Region), the draft permit, the August 2012 Application from Trinity Consultants (Application), and other materials in the record.¹

The CCL Project is proposed to be located in San Patricio and Nueces Counties, Texas. The liquefaction portion of the CCL Project consists of three ConocoPhillips Optimized CascadeLNG liquefaction trains, each capable of processing up to approximately 700 MMscf/d of natural gas with average liquefaction capacity of approximately 4.5 million tonnes per year. (Application at App. C.) Each liquefaction train includes a thermal oxidizers (TO) and six natural gas combustion turbines (CT), for a total of three TOs and 18 CTs. The draft permit includes a permitted GHG emission rate for each the CTs of 8,041 lbs CO₂/MMscf of LNG produced with 2,638,818 tons CO₂ per year (tpy), and a permitted GHG emission rate for each of the TOs of 115,600 lbs CO₂/MMscf with 589,374 tpy (Draft Permit at 8.) The total plant-wide annual GHG draft permit limit is 3,413,185 tpy CO₂e.

¹ Sierra Club relied on the materials posted at the following website: <http://yosemite.epa.gov/r6/Apermit.nsf/AirP>

The CCL Project is subject to GHG PSD regulations. New construction projects that are expected to emit at least 100,000 tpy of total GHGs on a carbon dioxide equivalent (CO₂e) basis, or modifications at existing facilities that are expected to increase total GHG emissions by at least 75,000 tpy CO₂e, are subject to PSD permitting requirements even if they do not significantly increase emissions of any other PSD pollutant. The Applicant estimates that the CCL Project will potentially result in a GHG emission increase of 3,413,185 tons per year (tpy) of carbon dioxide equivalents (CO₂e). (Draft Permit at 10.) The project would emit increased GHGs at a rate far greater than 100,000 tpy CO₂e; therefore, the project is subject to PSD review for all pollutants emitted in a significant amount. The Texas Commission on Environmental Quality (TCEQ) has assumed permitting responsibility for all non-GHG pollutants emitted from the proposed addition to the CCL Project. The Region's draft permit and these comments address only GHG related issues.

The CCL Project is one of dozens of proposed LNG export facilities in the United States.² Relatively cheap domestic natural gas price and higher global prices are creating a rush to market for LNG exports. However, this rush will have serious GHG consequences if it is not done responsibly and in a manner that realizes achievable GHG controls. The EPA has a clear mandate to act on climate change. The Application and supplemental response material demonstrate that this plant offers a clear opportunity to act on climate change by requiring an emission limit based on a determination that carbon capture and sequestration (CCS) is the best available control technology (BACT). Early LNG projects such as CCL and Freeport will set a precedent for the level of GHG controls that is achievable at the dozens of facilities to follow. It is imperative that the Region acts to ensure that facilities such as the CCL Project implement controls that will reduce climate changing greenhouse gases.

Texas is very vulnerable to climate changes and the Region must consider climate change impacts from the increased CO₂ emissions that would result from the CCL Project. Texas suffered its driest year ever in 2011, and the three years 2011-2013 have been among the driest on record. Cities are struggling to keep reservoirs full, and the Texas coast is experiencing accelerating sea level rise. Places like Galveston Island are spending substantial sums of money to keep the Gulf of Mexico at bay. The overriding environmental impacts of rising GHG emissions have been amply demonstrated. When considering the economic impact of GHG controls, the Region must acknowledge that the substantial environmental impacts from GHG emissions that Texas is facing, such as adverse impacts to water supplies, rising sea levels, and permanent shifts in existing agricultural ecosystems,³ override the economic elimination criterion in NSR review. (NSR Manual at B.32.)

² <http://energy.gov/sites/prod/files/2014/01/f6/Summary%20of%20LNG%20Export%20Applications.pdf>

³ R.T. Watson, M.C. Zinyowera, and R.H. Moss (Eds.), *The Regional Impacts of Climate Change: An Assessment of Vulnerability*, Cambridge University Press, 1997; C.B. Field and others, *Managing the Risks of Extreme Events and Disasters to Advance Climate Change Adaptation*, Cambridge University Press, June 2012; First through Fourth IPCC Assessment Report: *Climate Change 2007*, 2001, 1995, 1990. See also many other reports on the Intergovernmental Panel on Climate Change (IPCC) website at: http://www.ipcc.ch/publications_and_data/publications_and_data_reports.shtml#SREX.

I. INTRODUCTION

The proposed CCL Project is not designed to capture or otherwise control GHG emissions. The project relies on 18 simple-cycle aero-derivative turbines to power the LNG trains. CTs have much lower GHG energy efficiency than other alternatives such as combined cycle units or electric motors. The proposed CCL Project also does not include any reasonable attempt to capture the high purity CO₂ streams that are stripped out of the natural gas feedstock as part of the process to sweeten the natural gas for LNG processing. These flaws, (1) reliance on 18 CTs instead of electric compression and (2) ignoring a high purity CO₂ stream, result in a project that emits far more GHGs than could otherwise be achieved with an optimally designed facility.

Permitting for air pollutants must have meaning. The purpose of BACT is to require emission limits that reflect the best that the applicant can do to reduce harmful pollutants. These statements apply to GHGs just as they apply to other pollutants. Carbon capture and sequestration ("CCS") is a process that uses a chemical or physical solvent to remove CO₂, the dominant GHG, from a CO₂-containing stream (such as natural gas, flue gas, synthesis gas) using absorption, with subsequent stripping of the absorbed CO₂ to produce a concentrated CO₂ stream. Depending upon the acid gas removal technology applied, the CO₂ may need to be dried, then compressed to a dense phase state for pipeline transport to an appropriate storage location, most likely underground in a geological storage reservoir such as a deep saline aquifer or an oil reservoir or coal seam.

CCS is far and away the most effective technology available to reduce GHGs from industrial facilities like CCL, and it is the only potential add-on technology that the Region considered. Yet CCS has never been determined to be BACT. Here, the Region rejected CCS as technically infeasible on the basis that the add-on technology is not available for CTs. This conclusion ignores the fact that CCS is a proven technology, and it ignores the fact that a process separate from the combustion of fuel in the CTs (i.e. stripping CO₂ from feed gas) provides a much more economic opportunity to capture CO₂ from the CCL plant.

If the Applicant and the Region considered installing CCS for the Amine system alone, the estimated cost to control CO₂ would be \$35-40 per tonne. Compared to other recent BACT application estimates for the cost to control CO₂, these estimates strongly indicates that CCS is economically feasible for at least part of the CCL Project. The Region must therefore require a BACT limit for CO₂ based on the implementation of CCS on the Amine Units.

The Region must also consider replacing some or all of the 18 CTs with electric motors. The Region has already acknowledged in another recent BACT permit for the Freeport LNG facility that electric motors are an available and feasible alternative to on-site natural gas combustion to power the LNG trains. However, the alternative of using electric motors in lieu of CTs was not even considered in the draft permit or SOB. Instead, the Region states without explanation that "[t]he selection of compressor drivers was limited to gas fired turbines." (SOB at 10.) This omission is inappropriate given that the Region and the Applicant were fully aware that electric motors could feasibly provide a lower polluting alternative.

The Amine Units and TO's by themselves (i.e. without considering the CTs) would result in more than 552,000 tpy of CO₂ emissions. Reducing those emissions by substantially more than 99% and selecting optimal technology for acid gas removal, CO₂ drying, and compression will cost less than \$38 per ton. (Sierra Club Exhibit 1.) Since the CCL Project must include amine capture units to prepare the natural gas for liquefaction, the only added cost for CCS would be to

dry and compress the CO₂ and transport to an EOR field near Houston. At an annual operating cost (including capital recovery) of less than \$20 million, adding CCS would cost less than about 0.3% of expected total revenues from the LNG project, assuming annual revenues of \$5 to \$7 billion. In addition, replacing some or all of the 18 CTs with electric motors is technically feasible and would result in lower emissions of GHGs and other pollutants while increasing production of valuable LNG for sale.

II. THE REGION'S BACT ANALYSIS IS FUNDAMENTALLY FLAWED

The Region failed to conduct a thorough BACT analysis. Several potential control technologies were not even identified by the Region in Step 1 of the top-down BACT analysis. At a minimum, the Region must conduct an analysis of the control technologies discussed below and determine whether they constitute BACT. The record as it stands today is inadequate to determine BACT for the CCL Project because the Region did not address several of the most promising alternatives for controlling GHGs.

The Clean Air Act and U.S. EPA's implementing regulations require BACT emission limits for all new and modified pollution sources. 42 U.S.C. § 7475(a)(4); 40 C.F.R. § 52.21(j)(2). BACT is a limit based on the maximum degree of reduction achievable through, among other options, add-on controls. 42 U.S.C. § 7479(3) ("best available control technology" means an emission limitation based on the maximum degree of reduction of each pollutant... achievable for such facility through application of production processes and available methods, systems, and techniques, including fuel cleaning, clean fuels, or treatment or innovative fuel combustion techniques for control of each such pollutant"); *accord* 40 C.F.R. § 52.21(b)(12) (similar regulatory definition of BACT). The plain meaning of "maximum" is "the greatest quantity, number, or degree possible or permissible; the highest degree or point (of a varying quantity...) reached or recorded; upper limit of variation." *Websters New World College Dictionary* 837 (3rd Ed. 1997). Courts have thus instructed that the words "maximum" and "achievable" constrain the permitting agency's discretion in setting limits. *See Alaska Dept. of Env'tl. Conservation v. EPA*, 540 U.S. 461, 485-89 (2004).

The Environmental Appeals Board has repeatedly instructed permitting authorities that "BACT determinations are one of the most critical elements in the PSD permitting process, must reflect the considered judgment on the part of the permit issuer, and must be well documented in the administrative record." *In re Mississippi Lime Co.*, 15 E.A.D. __, PSD Appeal No. 11-01, Slip Op. at 17 (EAB, Aug. 9, 2011) (citing *In re Desert Rock Energy Co., LLC*, PSD Appeal Nos. 08-03 thru 08-06, slip op. at 50 (EAB, Sept. 24, 2009); *In re Knauf Fiber Glass, GmbH*, 8 E.A.D. 121, 132 (EAB 1999); *In re Newmont Nev. Energy Inv., LLC*, 12 E.A.D. 429, 442 (EAB 2005); *In re Gen. Motors, Inc.*, 10 E.A.D. 360, 363 (EAB 2002)). The result is a limit set based on the maximum achievable emission reduction with the best pollution control option that is "tailor-made" for that facility and that pollutant. *In re CertainTeed Corp.*, 1 E.A.D. 743, 747 (Adm'r 1982); NSR Manual at B.2 ("The reviewing authority then specifies an emissions limitation for the source that reflects the maximum degree of reduction achievable for each pollutant regulated under the Act.").

The list of control option types that must be considered when establishing a BACT limit includes both "add-on" controls that remove pollutants from a facility's emissions stream, and "inherently lower-polluting process or practices that prevent the pollutants from being formed in

the first place. *In re Knauf Fiber Glass*, 8 E.A.D. at 129. The New Source Review Workshop Manual describes the categories as follows:

Potentially applicable control alternatives can be categorized in three ways:

- **Inherently Lower Emitting Processes/Practices**, including the use of materials and production processes and work practices that prevent emissions and result in lower “production specific” emissions; and
- **Add-on Controls**, such as scrubbers, fabric filters, thermal oxidizers and other devices that control and reduce emissions after they are produced.
- **Combination of Inherently Lower Emitting Practices and Add-on Controls**. For example, the application of combustion and post-combustion controls to reduce NO_x emissions at a gas-fired turbine.

Office of Air Quality Planning and Standards, U.S. EPA, *New Source Review Workshop Manual* at B.10 (Draft, Oct. 1990) (“NSR Manual”); *see, also, PSD and Title V Permitting Guidance for Greenhouse Gas* at 25 (March 2011) (“GHG Guidance”).

BACT is a site-specific determination resulting in the selection of an emission limitation that represents application of control technology or methods appropriate for the particular facility. Any major stationary source or major modification subject to PSD must conduct an analysis to ensure the application of BACT. (NSR Manual at B.1.) The Region, in this case, employed the NSR Manual’s recommended methodology known as that “top-down” method for determining BACT:

The top-down process provides that all available control technologies be ranked in descending order of control effectiveness. The PSD applicant first examines the most stringent – or “top” – alternative. That alternative is established as BACT unless the applicant demonstrates, and the permitting authority in its informed judgment agrees, that technical considerations, or energy, environmental, or economic impacts justify a conclusion that the most stringent technology is not “achievable” in that case.

(NSR Manual at B.2.) The first step requires the permitting authority to identify all “potentially” available control options. *Id.* at B.5. The second step is to eliminate “technically infeasible” options from the potentially available options identified at step 1. *Id.* at B.7. In step 3 of the top-down method, the remaining control technologies are ranked and then listed in order of control effectiveness for the pollutant under review, with the most effective alternative at the top. In the fourth step of the analysis, the energy, environmental and economic impacts are considered and the top alternative is either confirmed as appropriate or is determined to be inappropriate. *Id.* at B.29. Issues regarding the cost-effectiveness (i.e. \$/ton) of the alternative technologies are considered under step 4. *Id.* at B.31-.46. The purpose of step 4 of the analysis is to validate the suitability of the top control option identified, or provide a clear justification as to why the top

control option should not be selected as BACT. *Id.* at B.26. Finally, under step 5, the most effective control alternative not eliminated in step 4 is selected and the permit issuer sets as BACT an emissions limit for a specific pollutant that is appropriate for the selected control method. *Id.* at B.53; *see, generally, In re Prairie State Generating Co.*, 13 E.A.D. 1, 11 (EAB 2006).

In 2011, EPA issued its *PSD and Title V Permitting Guidance for Greenhouse Gas* (“GHG Guidance”) to assist permitting authorities in addressing PSD and Title V permitting requirements for GHGs. Section III of the GHG Guidance addresses the BACT analysis. (GHG Guidance at 17-46.) The GHG Guidance directs permitting authorities to “continue to use the Agency’s five-step ‘top-down’ BACT process to determine BACT for GHGs.” *Id.* at 17. The GHG Guidance also notes that carbon capture and sequestration (“CCS”) should be considered an available technology in step 1 of the BACT analysis, and any elimination of CCS requires full and detailed documentation in the record. *Id.* at 33-34 (“if the permitting authority eliminates [CCS] at some later point in the top-down analysis, the grounds for doing so should be reflected in the record with an appropriate level of detail.”) With respect to step 4, the GHG Guidance acknowledges that the costs of CCS may be high in some cases, but permitting authorities must nevertheless explain any rejection of CCS in a well-documented permit record. *Id.* at 42.

A. The Region’s BACT Analysis Failed to Identify and Analyze all Potential Emission Control Options

Step 1 of the top-down BACT analysis requires the applicant to identify all control options with potential application to the source and pollutant under evaluation. Potentially applicable control options include: (1) inherently lower-emitting processes/practices; (2) add-on controls; and (3) combinations of inherently lower emitting processes and add-on controls. (NSR Manual at B.10.) The CCL BACT analysis failed to consider all potentially applicable control options.

The BACT analysis for the CT’s included the following alternatives (SOB at 6):

- carbon capture and storage (“CCS”)
- low carbon fuel
- design energy efficiency
- operational energy efficiency.
- Post-combustion catalytic oxidation
- N₂O catalyst

For the TO units, the SOB identified the following:

- Carbon capture and storage (“CCS”)
- Design measures
- Good combustion and maintenance practices

These control options failed to even consider (1) the use of electric motors in lieu of gas fired turbines, and (2) installing CCS on the high-purity waste stream from the amine units. By failing to identify and analyze these issues in the SOB, the Region has failed to allow the public to fully consider and weigh in on these issues.

1. The Region Failed to Consider the Use of Electric Motors in Lieu of CTs

The Region did not consider the alternative of installing electric motors in lieu of CTs. Without explanation, the Region stated in its analysis of design energy efficiency that, “[t]he selection of compressor drives was limited to gas fired turbines.” (SOB at 10.) There is no mention anywhere in the SOB of electric drives. This is surprising considering that the Region just recently (December 2013) issued a draft permit and statement of basis for the proposed Freeport LNG project (PSD-TX-1302-GHG) with the following: “Freeport LNG has determined that electric motor primary drivers are the most energy efficient of the available primary driver alternatives for the Liquefaction Process. Electric motors, in comparison to other drivers, produce no GHG emissions...” (Freeport LNG SOB at 9.)

The omission of any mention of electric motors in the CCL Project SOB is particularly troubling because information in the record indicates that the Region specifically questioned CCL about electric motors. The Region noted in an email to the Applicant that more information was needed “on the basis for eliminating electric driven compressors in the turbine BACT analysis.” (Email from Aimee Wilson, May 2, 2013.) However, rather than providing more information in the record and including electric motors as part of the BACT analysis, the Region failed to even identify electric motors as a potential GHG control technology in Step 1 of the top-down BACT analysis.

a) Electric Motors are Feasible

The CCL Project could do away with compression turbines entirely by using electrical compression. Electrically driven compression, powered by the grid, is a feasible alternative. Electric motor drivers have more efficient turndown characteristics for variable output operations, they can be sized to allow more efficient design, they generate no waste heat, and are not affected by the weather. (Turbines lose efficiency as temperatures and humidity deviate from design, a major issue for the Corpus Christi area, where high temperatures and humidity are present much of the year.) For these and other reasons, electric compression generally has lower emissions of conventional and greenhouse gas pollutants, even when emissions associated with upstream electricity production are considered. Electric compression and refrigeration therefore must be considered in the BACT analyses. If necessary, CCL can contract for power supply from a combined cycle power plant to assure that air emissions associated with electric power usage at CCL are as low as possible.

Multiple foreign facilities use electrically driven compression in natural gas liquefaction units. There are at least two LNG export terminals that use electrically driven compression for their natural gas liquefaction. These are the Melkøya Island/Snohvit and Stavanger terminals in Norway. In addition, industry documents describe liquefaction facilities in Indonesia and Africa that use electrically driven compression. The Stavanger, Norway LNG export facility uses electrically driven liquefaction designed by Linde with an annual capacity of 300,000 tons of LNG. The project was completed and ready for production in August 2010. (Exhibit 3).⁴ As another example, Siemens described Statoil’s Melkoya Island, Norway LNG export facility as “the first ever all-electric natural gas liquefaction plant,” with four 65-megawatt variable-speed

⁴ Available at: http://www.linde-engineering.com/internet.global.lindeengineering.global/en/images/LNG_3_3_e_11_150dpi19_15617.pdf

drive systems provided by Siemens. (Exhibit 4.)⁵ Other documents also detail the benefits of electric driven LNG. (Exhibit 5 and 6.) These industry documents indicate that electrically driven compression for natural gas liquefaction is feasible because it is a demonstrated technology that has been deployed in existing projects.

In the U.S., at least three applications for construction of LNG export terminals propose to use electric driven compression: the Freeport LNG Development, Jordan Cove Energy Project, and Oregon LNG.

The Freeport LNG terminal recently permitted by the Region included electric motors. In a response to concerns raised by the Region, the Applicant argued that electric motors was not BACT for Freeport LNG because the facility is proposed to be located in a nonattainment area for ozone and therefore must meet LAER. This distinction is meaningless for Step 1 and Step 2 of the BACT analysis. While it is possible that the consideration of adverse economic impacts in Step 4 of the BACT analysis could eliminate a technology that is otherwise required as LAER at another facility, the Region cannot make that determination without considering the technology. The fact that electric motors exist on other facilities means that the Region must at least consider that option in the BACT analysis.

The Jordan Cove LNG terminal, which was approved by DOE today, was described as follows: “The four LNG liquefaction trains will be electric and thus, only fugitive VOC emissions are expected from that equipment.” (Exhibit 7.)⁶ Another project developer, LNG Development Company, LLC (d/b/a Oregon LNG), described its project as using “[e]lectric motors will drive the propane and MR compressors.” (Exhibit 8.)⁷ The fact that Freeport, Jordan Cove, and Oregon LNG have all proposed to use electrically powered compression in their liquefaction processes demonstrates that electrically driven compression is a feasible control technology. These three proposals are proceeding in parallel with CCL’s proposal, and they demonstrate that electrically driven compression is an available and applicable control technology in the context of a large LNG export project such as the one proposed by CCL.

The Applicant asserts that the existence of other projects using electric motors does not mean that there are not adverse energy or economic impacts from using electric motors at CCL. However, to the extent those impacts exist, the Region must consider them in Step 4 of the BACT analysis and fully document its reasons for accepting or rejecting electric motors as a control technology.

b) Electric Motors Do Not “Redefine the Source”

In a June 4, 2013 email to the Region, the Applicant attempted to justify its exclusion of electric motors by arguing that Step 1 need not identify electric motors because this would be “redefining the source.” The use of electric motors to control compressors is not a redefinition of the source. The CTs are one component of a large LNG processing facility. The input (feed gas)

⁵ <http://www.energy.siemens.com/us/pool/hq/energy-topics/venture/downloads/Siemens%20VSDS%20for%20Statoils%20Hammerfest%20LNG%20plant.pdf>

⁶ <http://elibrary.ferc.gov/IDMWS/common/OpenNat.asp?fileID=13264807>. The Jordan Cove plant includes an adjacent power-generation facility, but the compressors still are electric.

⁷ <http://elibrary.ferc.gov/IDMWS/common/OpenNat.asp?fileID=13276695>

and output (LNG) of the CCL Project would be the same regardless of whether the compressors are powered by onsite combustion turbines or electric motors that use the grid.

To determine whether a potential control option would redefine the source, the Environmental Appeals Board has required permitting authorities to examine first how the applicant defined the proposed facility's "end, object, aim, or purpose," in other words, "the facility's basic design" as described in the application and supporting materials. *La Paloma Energy Center, LLC*, 16 E.A.D. ___, 26 (2014)(citing *Prairie State*, 13 E.A.D. at 22). The Board has cautioned that permitting authorities should not simply dismiss alternative control options, such as cleaner fuels, as constituting redesign, thereby creating an "automatic BACT off-ramp" from further consideration of the option. *Id.* Permitting authorities must look at source-specific technical impediments that would prevent a cleaner fuel source.

In this case, there are no insurmountable impediments to using electric motors. Any space constraints that might exist would likely be relieved by removing some or all of the CT turbines. The Applicant alleged in its June 4, 2014 email that electric motors cannot be used because the CTs provide all of the heating needs for the facility. This logic is faulty on two counts. First, the Applicant's own email identified the technical solution: "If additional electric motors were substituted for the gas turbines, new and additional direct-fired heaters would be required at the plant to make up for lost heating service." (June 4, 2012 email from Andrew Chartrand.) This is not a valid basis for claiming that electric motors would "redefine the source." There are other potential sources of heat, including the installation of direct-fired heaters. The direct-fired heaters will have emissions, but those should be considered compared to the emission of the CTs that they would be replacing.

With respect to GHG emissions only, the Applicant asserts that the use of electric GHG motors would result in offsite GHG emissions from power generators on the grid. However, the Texas grid is, on average, much more efficient than a series of 18 simple-cycle CTs. Texas is one of the largest producers of emissions-free wind power, and Texas also has many baseload natural gas units that operate more efficiently than CTs. To the extent that the Applicant or the Region claim that reliance on the grid would increase offsite GHG emissions, they must provide some evidence showing that to be the case. The on-site evidence of electric motors versus CTs clearly shows that electric motors would be much cleaner. The Applicant's argument also cannot overcome the option of CCL contracting directly with an electric generator to provide low-emission power. If nothing else, a large combined-cycle natural gas unit would be more efficient than 18 CTs.

The Applicant's argument that it needs heat from the CTs is also faulty because it ignores the potential to replace only some of the 18 CTs with electric motors. The CCL Project design presented by the Applicant includes the installation of a waste heat recovery unit (WHRU) on 6 of the 18 combustion turbines. (SOB at 6.) The Applicant has asserted that this is sufficient to provide 100% of the heating needs of the facility. Therefore, the remaining 12 CTs could be replaced with electric motors without any impact on the heating needs of the facility. This alternative would still reduce two-thirds of the total emissions from the CTs. The Region must analyze this option to avoid providing an "automatic BACT off-ramp" to the CCL Project and other LNG applicants.

c) The Region Must Provide the Public an Opportunity to Consider and Comment on a Full BACT Analysis for Electric Motors

Sierra Club is concerned that the Region did not present its justification for considering and rejecting electric motors in the SOB and draft permit. Obviously the issue of whether electric drives could or should be used was an issue of concern for the Region, as evidenced by the email exchanges between the Region and the Applicant. “It will be extremely difficult for us to defend eliminating electric drive compressors for LNG liquefaction when that is what is proposed by Freeport.” (May 2, 2014 Email from Aimee Wilson.) The Region’s question was valid, and the information provided in response could have been incorporated into a BACT analysis of BACT. But the Region did not “eliminate” electric motors; it never even identified electric motors. Rather than addressing the issue in the SOB, the Region appears to have tried to avoid the issue altogether by simply ignoring the possibility that electric motors could be used. This raises the concern that the Region is only interested in the information on electric motors because it is (correctly) concerned that the issue would be raised in public comments. Sierra Club expects based on past permitting decisions that the Region will simply respond to the issue of electric motors for the first time in its Response to Comments issued with the final permit. However, that approach is harmful to public participation because it does not allow Sierra Club or other members of the public to comment on the Region’s analysis of the issue.

The email exchange identified by Sierra Club here does not overcome this problem because the emails do not present any indication of the Region’s assessment of the issue beyond raising the original question. Unless Sierra Club is to assume that the Region agrees with CCL on the issue without question, which as discussed above would be problematic and inconsistent with BACT, then Sierra Club and other members of the public cannot provide a fully informed comment on the issue. There is no excuse for the Region withholding its assessment of electric motors in the SOB after it has already identified that issue as a concern. The Region must remedy this deficiency by first issuing a new draft permit and SOB that includes a full BACT analysis of electric motors and then providing an additional opportunity for public comment.

2. The Region Failed to Consider CCS on the Amine Units

The Region failed to identify installing CCS on the high-purity CO₂ stream from the amine units prior to the TOs as a potential control alternative. This type of design and control is both feasible and far more economical than installing CCS on post-combustion waste streams from either the TOs or the CTs.

The most effective control for GHGs, CCS, is capable of removing 90%+ of the CO₂ and thus, if it is found to be cost effective and without other adverse impacts, would qualify as BACT. The proposed design now already captures almost all of the CO₂ contained in the natural gas feed. This CO₂ must be removed to prevent CO₂ hydrates from forming in the liquefaction trains. However, CCL and the Region rejected CCS as technologically infeasible for both the CTs and the TOs based on the assumption that the waste stream from both units would be combined and would therefore be a low-purity concentration of about 3.5 percent CO₂. (SOB at 7.) By combining high purity CO₂ from the TO’s and with very low purity CO₂ from the CTs, CCL has foreclosed on the most attractive CCS opportunity.

The Region applied the same rationale for rejecting CCS for the TOs as it did for the CTs. (SOB at 17.)⁸ However, the Region assumed that CCS on the TOs could only be accomplished using post-combustion controls similar to a natural gas fired turbine. This conclusion ignored the fact that the CCL Project will already result in a high-purity CO₂ stream from the amine units that are designed to strip out CO₂ from the feed gas. Rather than simply sending that stripped out CO₂ to the TO units to later be mixed with exhaust from the CTs, the Applicant should have considered whether it was feasible to capture this high purity CO₂ stream and store it before mixing it with the lower-purity stream from the CT units. Diverting and capturing the high-purity CO₂ stream would be more economical than trying to capture the same CO₂ after it had been diluted with the exhaust of the CT units. The CO₂ leaving the amine units is free of sulfur compounds and can be fed to a CO₂ compression and drying facility for sale to enhanced oil recovery (EOR) operators, or it could simply be injected for permanent geological storage.⁹ If CCL captured this stream of CO₂, it most likely would not even need to install the expensive TO's, thus saving the project millions of dollars. The Region failed to even consider this control alternative for the Amine units before the exhaust stream was sent to the TOs, and therefore the draft permit is deficient.

The Region clearly erred by failing to identify capture of the high-purity CO₂ stream from the amine units prior to combustion in the TOs. This is either a Step 1 problem (failure to identify control alternatives) or a Step 2 problem (improperly rejecting CCS as technically infeasible). In either case, the Region must conduct a proper BACT analysis that at least considers CCS for the amine units at CCL in Step 4 of the top-down BACT analysis.

a) Sierra Club Estimated Costs of CCS on Amine Units at CCL

The current record is inadequate to conduct a thorough economic analysis that would be required to reject CCS in Step 4 of the BACT analysis. In CCL's Application, it included an analysis of CCS for the TO CO₂ (Application at Appendix D). CCL based its analysis of CCS costs CCS estimates from a 2010 DOE/NETL study of gas and coal fired power plants with post combustion CO₂ capture. CCS simply plugged in its output emission numbers to the DOE/NETL study and assumed everything else would be the same. This analysis is patently incorrect as applied to the CCL facility because CCL already has an amine unit to strip out CO₂ from the natural gas feed. The 2010 DOE/NETL estimates relied on by CCL assumed post-combustion capture from natural gas plants or coal plants. This means that a large portion of the cost estimates from that study assumed the need to build an amine unit. It also assumed that the amine unit would need to be sized to process a lower purity CO₂ stream that would come from a coal or gas combined-cycle combustion turbine.¹⁰ The CCL analysis, and the Region's wholesale

⁸ Sierra Club also disputes the Region's conclusion that CCS is technically infeasible for the CTs. See below for more detail.

⁹ The coastal plains region in Texas along the Gulf Coast contains 65% of the country's estimated accessible carbon storage resources, with an estimated 2,000 gigatons of accessible storage resources. U.S. Geological Survey Geologic Carbon Dioxide Storage Resources Assessment Team, 2013, National assessment of geologic carbon dioxide storage resources—Results: U.S. Geological Survey Circular 1386, p. 41. Available at: <http://pubs.usgs.gov/circ/1386/>

¹⁰ CCL also erred by asserting in Appendix D of the Application that a "high" and "low" cost estimate for CCS was included in the DOE/NETL study. CCL used the "average" of these estimates. In fact, the "high" estimate in the DOE study is for a coal plant. The "low" estimate is for a natural gas combined-cycle turbine. The actual cost of

adoption of that analysis, fails to meet the BACT requirement to perform a source-specific analysis. The CCL cost estimate did not even consider the same type of industrial process as the 2010 DOE/NETL study.

The assumptions made in the DOE/NETL are inappropriate for the CCL Project. Although the CCL amine unit may have to be optimized for recovery of CO₂ for EOR, or possibly include a solvent based system to be used to avoid the drying step, the facility is already designed to use the amine CO₂ for beneficial use. CCS at the CCL Project would simply mean diverting the already-captured high-purity CO₂ stream before it entered the TOs. CCL could actually save money because the TOs would likely be unnecessary.

Notwithstanding the deficiencies in the record, Sierra Club developed a cost estimate based on similar methods from the Freeport LNG GHG PSD permit application (conducted by Atkins). (PSD-TX-1302-GHG)(Exhibit 2).¹¹ Sierra Club Exhibit 1 is a spreadsheet that added the CCL amine unit outputs into the assumptions used by Atkins for the Freeport LNG application. Sierra Club then removed costs for building an amine unit (which already exists at the CCL Project) and estimated the added net capital cost for compressing, drying and transporting the amine CO₂ from the CCL Project to the Houston Area (assuming the Denbury Hasting Field as the destination, as assumed in both the Freeport and CCL PSD BACT Studies).¹² Aside from building a pipeline to Denbury, the only added cost to add CCS to the CCL Project would be for the CO₂ compression and drying equipment. For these estimates, Sierra Club used updated 2013 estimates from the DOE CCS Study (DOE/NETL-2013/1614), adjusting for scale and escalation to 2014. We estimate the annual cost to deliver CO₂ to the Hastings area would be about \$20 million per year, or less than \$38/ton. If the EOR operation would pay any positive price for the CO₂, the annual cost to the CCL Project would be even lower. CCL and the Region should also investigate whether EOR operators would pay some or all of the pipeline construction and transportation costs. If oil prices rise, delivering CO₂ from the CCL Project could actually generate positive cash flow for the CCL Project.

The bulk of the costs leading to Sierra Club's estimate of \$38/ton to control CCS from the CCL Project amine units come from pipeline cost. This cost could be reduced or eliminated if CCL or the Region identified a closer source for geological sequestration. As noted above, the Gulf Coast is a particularly attractive area for CO₂ storage opportunities. The Region must thoroughly investigate these options as it develops its CO₂ storage estimates.

B. The Region Improperly Determined that CCS on the CTs Is Technologically Infeasible

The Region never reached a cost analysis in Step 4 of the BACT analysis for CCS at the CCL Project because it improperly conclude in Step 2 that CCS was technologically infeasible for simple-cycle units. (SOB at 8-9.) The Region based this conclusion on a single feasibility study for a proposed Alaskan gas processing plant known as the Central Gas Facility (CGF). The

CCS for a high-purity CO₂ stream would be much lower than both of these estimates. CCL's use of an "average" cost is completely unjustified.

¹¹ Although Sierra Club did not agree with all aspects of the Freeport LNG analysis, and commented accordingly, the analysis performed by the Freeport applicant's consultant, Atkins, was far more detailed and much better than the analysis performed by Trinity in the CCL application.

¹² Sierra Club's estimate of cost for a 160 mile pipeline was about the same as CCL estimated.

Region apparently concluded that, based on the CGF feasibility study, CCS at the CCL Project was infeasible for the 18 CTs because heat (steam) would be required to operate a proposed MEA unit. (SOB at 9.) In Alaska, the authors of the CGF feasibility study concluded that no heat would be available from the simple cycle units because it would require a heat recovery steam generator. CGF therefore rejected CCS because “it would essentially require conversion of the turbines from simple-cycle to combined-cycle.” (SOB at 9.)

This logic is inapplicable to the CCL Project. First, the CCL Project already includes the installation of a waste heat recovery unit (WHRU) on six of the eighteen combustion turbines. (SOB at 6.) The Applicant explained that these WHRUs will “provide all the heat required for regenerating the amine used in the acid gas removal unit, to regenerate the molecular sieves used to dehydrate the feed gas, and to provide heat in the heavies removal unit.” (Jan. 31, 2014 Email from Andrew Chartrand, Comment CSV3) (emphasis added). In other words, the CCL project is already capturing waste heat from six of the CTs to power the amine unit that it already needs to process the natural gas. This means that the remaining twelve CTs could be used to produce more heat to power an MEA CCS unit for a waste stream from the CTs. However, the Applicant rejected even attempting to capture any more heat from those CTs: “The only conceivable use for steam generated by a HRSG installed on the other gas turbines would be to install steam turbines to generate electrical power.” (*Id.* at Comment ahb7.) The Applicant is wrong. Another conceivable use for steam generated by a HRSG or WHRU could be to power a CCS system for the waste stream from the CTs.¹³

Second, the Region failed to consider that a separate boiler could be installed to provide steam heat to operate the MEA. Construction of such a boiler has been included as part of the costs of installing a CCS system in previous BACT analyses. Such an addition would add capital costs to the CCS system, but those issues are appropriately addressed in Step 4 of the BACT analysis. In any case, building a new boiler would be unnecessary in this case because the Applicant could simply add WHRUs to some or all of the remaining CTs.

In short, CCS is technologically feasible for the CTs. Rejecting CCS as technologically infeasible in Step 2, as the Region has done here, ignores the fact that designing a CCS system for the CTs is achievable. In fact, the Applicant essentially already designed such a system because it is powering existing amine units to strip out CO₂ from the natural gas feed before processing it. A similar routing of excess steam could be designed and the steam used to power an amine unit to strip out CO₂ from the CT waste streams. Such a system would likely cost more per ton of CO₂ removed than capturing CCS from the high-purity waste stream from the amine units discussed above, but the cost of that process must be considered in Step 4 rather than rejected in Step 2 of the BACT analysis.

¹³ Sierra Club also notes that, as discussed above, the CCL Project already includes amine units to treat the natural gas feed. In fact, powering those amine units is one of the reasons given by the applicant for the WHRU that are already planned for the project.

C. CCL Excluded Design Optimization For CCS That Would Have Resulted In Significantly Lower Costs.

The definition of BACT includes the “application of production processes or available methods, systems, and techniques, including fuel cleaning or treatment or innovative fuel combustion techniques.” CCL did not consider “available methods, systems, and techniques, including fuel cleaning or treatment or innovative fuel combustion techniques” in Step 1 of its BACT analyses.

Many of the processing steps for CCS could be eliminated for the CCL Project by simply designing the CCL facility to recover CO₂ for EOR or geological sequestration, rather than attempting to “bolt-on” a control design after the fact. This optimized design change would involve, for example, replacing the amine units with a solvent-based system such as Selexol. The CO₂ stripped from the solvent would be low water content and sulfur free. It could then go directly to a compression drying plant and then an EOR pipeline such as Denbury, eliminating or reducing most of the drying cost, additional sulfur polishing, the blower, refrigeration, and injection pumps that CCL included in its cost analysis.

The starting point of any cost analysis is the battery limits and design basis. (NSR Manual at B.32 - B.33.) The battery limits are the area or process segment to be costed. In this case, the project purpose of the CCL facility is to produce treated gas suitable for the downstream liquefaction and export of LNG. The goal in the BACT analysis should therefore be to select a control alternative that minimizes GHG emissions and other pollutants. In this case, the means creating a high purity CO₂ stream suitable for CCS or CCS/EOR, rather than to reduce a pollutant, such as NO_x or SO₂, with add-on controls. Thus, the battery limits for CCS and CCS/EOR should include the design of the entire CCL facility. Considering alternative designs for the CCL Facility does not change the overall project purpose, which is to treat natural gas for delivery to the LNG export terminal.

The design basis used to cost CCS and CCS/EOR was not disclosed, as discussed elsewhere in these comments. However, inspection of the cost analyses indicates that CCL Facility design was not optimized for CO₂ recovery, thus artificially raising the costs. CCL’s CCS and CCS/EOR cost analyses are based on the original design to upgrade natural gas for liquefaction, without any consideration of CO₂ capture and use. This is a fundamental and serious engineering error.

The CCL cost analyses assume a static design based on the amine process used to upgrade natural gas for liquefaction. The facility design was not optimized to consider the capture and use of CO₂. However, in standard engineering practice, this is not how a CCS or CCS/EOR system would be designed. The design basis, or starting point, for the entire CCL Facility would assume the end use of the CO₂. The design basis of the CCL Facility should have been the best design for producing a high purity CO₂ stream suitable for CCS and CCS/EOR at the lowest cost. The BACT analysis is therefore flawed in Step 1 because it failed to consider alternative processes in the CCL Facility capable of delivering a higher purity CO₂ stream suitable for CCS or CCS/EOR.

If the CCL Facility were designed with the goal of capturing the CO₂ for geological sequestration, EOR or other end uses, an amine system would not have been selected to pretreat the natural gas for liquefaction because it generates a CO₂ stream that requires more costly treatment to meet downstream requirements. Rather, a selective acid gas removal (AGR)

technology that uses a physical solvent would have been selected. A 2-stage Selexol (UOP)¹⁴ or Rectisol (Lurgi, Air Liquide, Linde) unit,¹⁵ for example, could selectively remove both CO₂ and sulfur compounds to EOR specifications. The resulting CO₂ stream would have a low water content and a lower sulfur content and could go directly to a compression and smaller drying plant and then to a pipeline. These systems would eliminate the sulfur treatment costs, avoid most of the drying costs, eliminate the blower, refrigeration, and additional injection pumps.¹⁶ Currently proposed commercial CCS systems (e.g., HECA, Summit) are based on either Selexol or Rectisol. As of 1992, 12 Selexol plants had been installed to remove CO₂ from natural gas.¹⁷ Other similar, recently built and proposed projects also use Selexol. The giant Sandridge natural gas plant in West Texas, for example, is currently recovering 5 tonne/yr of CO₂ for EOR using the Selexol process. A second train, with a capacity of 3.4 tonne/yr of CO₂ was completed in late 2012.¹⁸

These examples show that a more efficient system design is available for the CCL Facility. Requiring an analysis of such a system design would not change the overall project purpose because the end result, treatment of natural gas for the downstream Liquefaction Plant, would not change. Neither CCL nor the Region considered other alternative for the CCL Facility design. The Region must require a holistic review of the Pretreatment Facility and consider whether an alternative design that optimizes treatment of the natural gas for CCS would be a cost-effective method of controlling GHGs.

D. CCL Failed To Evaluate Other Treatment Options

The bolt-on approach used by CCL only considered a single processing train: gas treatment/dehydration, compression, and sulfa (or Triazine) treatment. CCL's cost analyses of all of the options failed to consider other methods to reduce CO₂. These include other downstream capture methods and alternate fuels. The definition of BACT specifically includes the "application of production processes or available methods, systems, and techniques, including fuel cleaning or treatment or innovative fuel combustion techniques."

First, CCL only considered one processing train design to capture CO₂ emissions, based on additional amine and sulfa treatment. There are other options that should have been evaluated. These include the use of ExxonMobile's Controlled Freeze Zone technology, which is in use at the LaBarge facility's CO₂ capture plant.¹⁹

¹⁴ UOP Selexol™ Technology for Acid Gas Removal, 2009. Available at: <http://www.uop.com/?document=uop-selexol-technology-for-acid-gas-removal&download=1>.

¹⁵ Arthur L. Kohl and Richard B. Nielsen, Gas Purification, 5th Edition, Gulf Publishing Company, Chapter 14: Physical Solvent for Acid Gas Removal, 1997.

¹⁶ Kohl and Nielsen 1997, pp. 1202-1209. See, for example, Table 14-10, Plant B, which treats a natural gas similar to CCL.

¹⁷ Kohl and Nielsen 1997, p. 1203.

¹⁸ Global CCS Institute, Century Plant, Available at: <http://www.globalccsinstitute.com/project/century-plant>

¹⁹ LaBarge Fact Sheet: Carbon Dioxide Capture and Storage Project. Available at: http://sequestration.mit.edu/tools/projects/la_barge.html; ExxonMobile, Controlled Freeze Zone Increasing the Supply of Clean Burning Natural Gas.

Second, the combustion of natural gas in the CTs generates 2,638,818 ton/yr of CO₂e. (Draft Permit at 10.) The resulting gas stream has a very low CO₂ concentration and high nitrogen content, resulting in relatively high costs for CCS for the CT exhaust gases. The combustion turbine CO₂e emissions could be eliminated by replacing the natural gas with hydrogen. Hydrogen-fueled turbines are commonly used in IGCC projects. Hydrogen has no carbon and thus produces no CO₂ or methane.

The hydrogen could be obtained from a hydrogen pipeline system. Steam methane reforming produces a highly concentrated CO₂ stream, which can be more cost effectively upgraded for CCS or CCS/EOR than turbine exhaust. Air Products, for example, recently retrofitted its two Port Arthur SMRs with a vacuum swing adsorption (VSA) system to separate up to 4 million tons/yr of CO₂ from the process gas stream, followed by compression and drying.²⁰ It is currently delivering about 1 million tons/yr of CO₂ via a pipeline owned by Denbury for EOR.²¹

The Air Products SMR CCS project started up in May 2013. They feed hydrogen into Air Products' hydrogen pipeline supply network, the world's largest system of its kind, which stretches 600 miles from the Houston Ship Channel in Texas to New Orleans and is fed by over 20 Air Products' hydrogen production facilities and provides over 1.2 billion cubic feet of hydrogen per day to refinery and petrochemical customers.²² CCL could purchase hydrogen from this existing system and eliminate the hydrogen's GHG footprint by financing the retrofit of a producer SMR facility with CCS/EOR at a net cost of nearly zero.

The Corpus Christi area does not now have a hydrogen pipeline network like the one in the Houston refining centers. However, given the scale of the operation of CCL, an industrial gas company might be interested in construction an SMR to serve CCL and area refineries.

The Region must consider these alternative configurations for the CCL Facility as part of the BACT analysis. These changes, if found to reduce GHG emissions, would not constitute a change in the project purpose because there would be no change to the input or output of the CCL Facility. The purpose of the CCL Facility is to treat natural gas to remove CO₂ and other impurities before sending it to the terminal for export. Alterations to the method of how that natural gas is treated and any associated emissions reductions would not change the basic project purpose for CCL. Therefore, design changes to the CCL Facility must be considered in step-1 of the BACT analysis.

²⁰ Air Products and Chemicals, Inc: Demonstration of CO₂ Capture and Sequestration of Steam Methane Reforming Process Gas Used for Large-Scale Hydrogen Production, Available at: <http://www.netl.doe.gov/publications/factsheets/project/FE0002381.pdf>

²¹ Global CCS Institute, Air Products Steam Methane Reformer EOR Project, Available at: <http://www.globalccsinstitute.com/project/air-products-steam-methane-reformer-eor-project>

²² Air Products, Air Products Celebrates Texas Carbon Capture Demonstration Project Achievement, New Release, Available at: <http://www.airproducts.com/company/news-center/2013/05/0510-air-products-celebrates-texas-carbon-capture-demonstration-project-achievement.aspx>.

Sierra Club appreciates the opportunity to provide these comments.

Sincerely,

/s/ Travis Ritchie

Travis Ritchie

Associate Attorney

Sierra Club

85 Second Street, Second floor

San Francisco, CA 94105

(415) 977-5727

travis.ritchie@sierraclub.org